Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

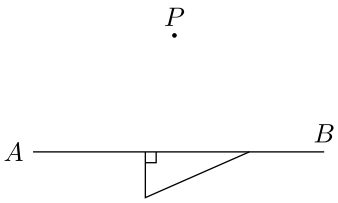
Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing a Line Parallel to Another

Two straight lines are parallel when they never meet, however far you draw them, and stay the same distance apart all along their length. In this construction you are given a straight line AB and a point P, and your job is to draw a second line through P that is parallel to AB — a line that runs alongside AB, the same distance away at every point, never crossing it.

You will need a set square and a ruler. The set square has a right angle in one corner, and it is that corner (marked in the figures) that makes the construction work. Study fig-01: the line AB lies flat across the page, with the point P marked above it. Your first task is to line up the set square with AB, and the right angle is the part that keeps it true.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Construct a line through P parallel to AB.

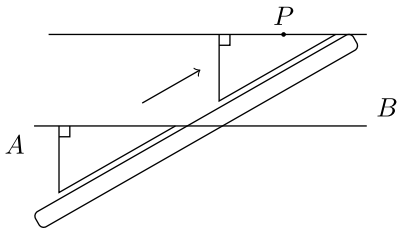
(a) Place the set square so that one of its straight edges lies accurately along the line AB, as in fig-01.

(b) Put the ruler against one of the other edges of the set square — the left-hand edge if you are right-handed. In fig-02 the ruler lies along the set square's downward edge; in fig-03 it rests against the sloping edge instead. Which edge you choose does not matter: the only thing that matters is that the ruler is held firmly against the set square and does not move once it is in place.

(c) Hold the ruler firmly and slide the set square along it towards P. Keep sliding until the edge that was lying along AB reaches P. Then draw a line along that edge of the set square, through P — this is shown in fig-04 and fig-05.

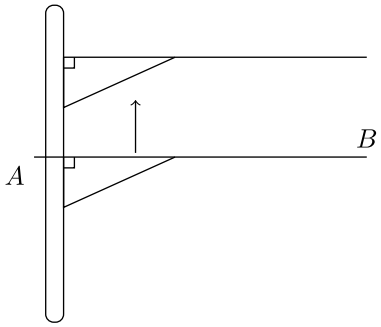
Why does this work? Sliding the set square along the held ruler moves it without turning it, so the edge that lay on AB keeps exactly the same direction the whole way. A line drawn through P along that edge therefore runs parallel to AB, the same distance away all along — which is precisely the line you wanted, as fig-05 shows.

Visual



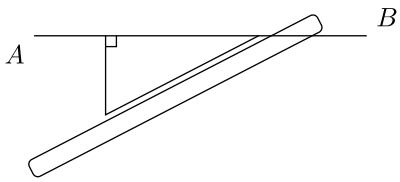
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



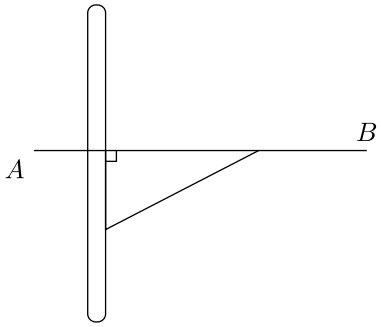
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Practice

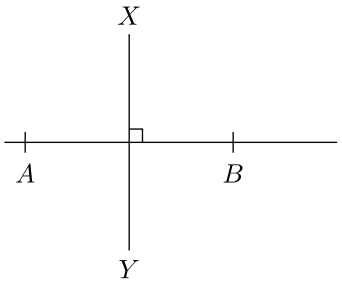
1. You are given a line AB and a point P above it. What is the first step in constructing a line through P parallel to AB?
2. After the set square is lying on AB, what do you do with the ruler before you slide the set square?
3. While sliding the set square along the ruler towards P, what must you be careful to do, and why does that keep the line you draw parallel to AB?

Section 2: Constructing Perpendicular Lines

Two lines are said to be perpendicular to each other when they meet at a right angle — that is, at an angle of $90\,^{\circ}$.

Look at fig-06. The line XY crosses the line AB, with X above and Y below. The small square drawn at the crossing marks the right angle in the upper right quadrant. Because XY meets AB at $90\,^{\circ}$, the two lines are perpendicular to each other.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Let us construct a line perpendicular to a given line, using a set square and a ruler.

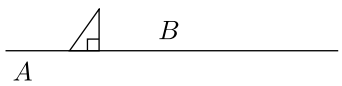
Step 1 — Draw the line and mark the point. Draw a line AB as in fig-07, and mark a point Q on the line where you want the perpendicular line to pass.

Step 2 — Place the set square. Place the edge of the set square on line AB, so it sits on the line as shown in fig-08. Now hold the ruler so that its edge meets the hypotenuse (the long sloping edge) of the set square, as the figure shows.

Step 3 — Slide and draw. Pressing firmly on the ruler so it does not move, slide the set square down along the hypotenuse until the third edge of the set square touches point Q. Then draw a line along that edge, from X above the line down to Y below it. Fig-09 shows the result: the line XY now crosses AB at a right angle, marked with the small square.

Because the third edge of the set square is perpendicular to its base, the line you drew is perpendicular to AB. The two lines meet at $90\,^{\circ}$, so $XY \perp AB$.

Visual



Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

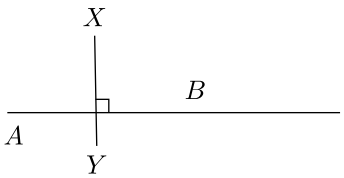
Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

The key point to remember: a line drawn perpendicular to another line meets it at a right angle, which is $90\,^\circ$. The set square has a built-in right angle, so sliding it along the ruler and drawing along its third edge gives you a line that is exactly perpendicular to your line.

Visual



Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Practice

1. When are two lines said to be perpendicular to each other?
2. Name the two instruments you use to construct a line perpendicular to a given line.
3. In the construction of a perpendicular line, what do you press firmly on so that it does not move while you slide the set square?
4. A line PQ is drawn so that it crosses line AB and makes an angle of $90\,^\circ$ with it. What is the relationship between PQ and AB?

Section 3: Bisecting a line segment

To bisect a line segment means to divide it into two equal halves. The point where the segment is cut into those two equal halves is called its midpoint. Using a pair of compasses and a ruler, you can find that midpoint exactly — and, as a bonus, the line you draw to do it meets the segment at a right angle (90°).

You need a mathematical set for this. Here is the whole procedure.

Step 1. Draw the line segment you want to bisect and label its ends, say A on the left and B on the right (fig-10).

Step 2. Open your pair of compasses so that the radius (the gap between the sharp point and the pencil) is about $\frac{3}{4}$ of the length of AB.

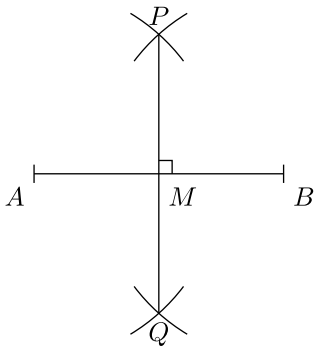
Step 3. Place the sharp point of the compasses on A. Draw two arcs: one above the middle of AB and one below the middle of AB.

Step 4. Keeping the same radius (do not change the compass opening), place the sharp point of the compasses on B. Draw two more arcs so that they cut the first pair of arcs — one crossing above AB and one crossing below. Label the upper crossing P and the lower crossing Q (fig-11). Each small X is a point where an arc from A meets an arc from B.

Step 5. Join P to Q with your ruler (fig-12).

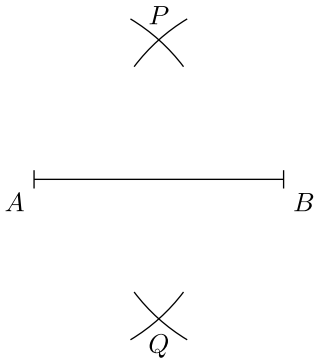
Step 6. The line PQ crosses AB at M. M is the midpoint of AB, so $AM = MB$, and PQ meets AB at $90\,^\circ$.

Visual



The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The line segment AB, a stroke at each end, A on the left and B on the right.

Worked example — bisecting the segment XY

Suppose you have a segment XY and you want to bisect it.

1. Open your compasses so the radius is about $\frac{3}{4}$ of XY.
2. Put the sharp point on X. Draw one arc above the middle of XY and one below it.
3. Without changing the compass opening, put the sharp point on Y. Draw two more arcs so they cut the first pair — one crossing above and one crossing below.
4. The upper crossing is P and the lower crossing is Q.
5. Join P to Q with a ruler. This line crosses XY at its midpoint M.
6. Read off the result: $XM = MY$, and PQ meets XY at 90°.

Practice

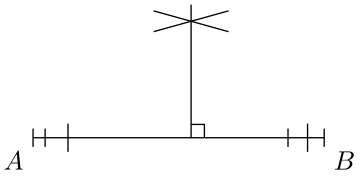
1. What does it mean to bisect a line segment AB? Where is the point where the bisecting line crosses AB?
2. You draw the two arcs from A with a certain compass opening. When you move the sharp point to B, what must you do with the compasses before drawing the second pair of arcs, and why?
3. After you join P to Q with a ruler, the line PQ crosses AB at M. What two facts do you know about M and about how PQ meets AB?
4. A segment PQ is 12 cm long. If you bisect it, how long is each half, and what angle does the bisecting line make with PQ?

Section 4: Constructing a 90° Angle

You can build a 90° angle with nothing but your ruler and compasses. The idea is to raise a perpendicular from a point on a straight line. Here is how it works, one step at a time.

1. Draw a straight line and mark a point C on it — that is the point where the 90° angle is to be built. In the figure (fig-13) the line runs from A on the left to B on the right, and C sits near the middle.
2. Put the point of your compasses at C and draw an arc that cuts the line at two points. Call those two points D and E, one on each side of C.
3. Now open your compasses WIDER than the width DC you used in step 2. This wider setting is essential: if you kept the old width, arcs drawn from D and E would only meet back at C itself, and that gives you nothing new. With the compasses at D at this new wider setting, draw an arc above the line.
4. Keeping the compasses at the same wider setting, place the point at E and draw another arc above the line, cutting the first arc at a point F.
5. Join C to F with your ruler.
6. The line CF is perpendicular to AB, so the angle between CF and the line, $\angle FCB$, is exactly 90°.

Visual



A line with a stroke at each end, A on the left and B on the right, and four short strokes standing on the line, a pair towards A and a pair towards B. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Let's watch the whole construction come together. In fig-13 you can see a line AB with a point C marked on it. Starting from C, an arc is drawn that cuts the line in two places — the four short strokes standing on the line show those two cuts, a pair towards A and a pair towards B; together they are the two points D and E.

Then the compasses are opened wider than the first width, and arcs are struck from D and E above the line. The two short strokes crossing each other near the top of the perpendicular show where those wider arcs meet — that is point F.

Finally a ruler joins C to F. The perpendicular rises from C, and the little square at its foot marks the right angle: $\angle FCB = 90^\circ$.

The key move to remember is step 3 — you must open the compasses wider than DC before drawing the arcs from D and E. With the old width the two arcs would only meet back at C, so you would never reach F.

Practice

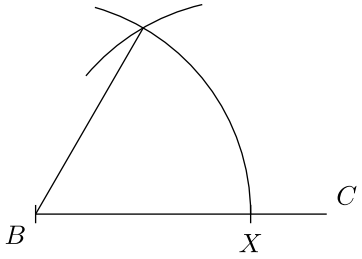
1. Write down the steps, in order, you would take to construct a 90° angle at a point C on a straight line AB.
2. In constructing a 90° angle at point C on a line, you first draw an arc with compasses at C to mark points D and E. Explain why you must open the compasses WIDER than DC before drawing the next arcs from D and E.
3. You have carried out the construction and joined C to F, where F is where the two wider arcs met. What can you say about the angle between CF and the original line AB? What is the size of $\angle FCB$?

Section 5: Constructing a 60° Angle

An angle of 60° is one you can build with just a ruler and a pair of compasses, using two arcs of the same radius. The key idea: when you swing an arc of a fixed radius from a point, every point on that arc is the same distance from the centre. Two arcs of the *same* radius from two different centres will meet at a point that is that same distance from both — and that equal spacing is what makes the angle 60°.

The construction needs just three moves. You can follow them on fig-14.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Constructing a 60° angle at B, along the line BC

1. **Draw the line.** Draw a line segment and mark a point B on it where the angle will sit. Mark the far end of the line C. Set your compasses to any convenient radius — it can be any size that fits comfortably on your page.
2. **First arc from B.** With the compass point on B, draw an arc that cuts the line BC at a point. Call that point X. (On fig-14 this is the long arc swung from B through X.)
3. **Second arc from X.** Keep the compasses at exactly the *same* radius — do not change the setting. Now put the compass point on X and draw another arc that cuts the first arc. Call the point where the two arcs cross A. (On fig-14 this is the shorter arc swung from X cutting the first.)
4. **Join A to B.** Draw a straight line from B to A.

The angle between line BC and line BA is exactly 60°. Because both arcs were drawn with the same radius, the distance from B to A equals the distance from B to X, which equals the distance from X to A — so the triangle BAX has all three sides equal, and a triangle with three equal sides has all three angles 60°.

Practice

1. Construct a 60° angle at the left-hand end of a horizontal line, following the steps above. Label the line, the first arc's cutting point, and the crossing point of the two arcs, then join up to complete the angle.

Check your work

Attempt every question before you read this page.

Constructing a Line Parallel to Another

1. Place the set square so that one of its straight edges lies accurately along the line AB.
2. Place the ruler against one of the other edges of the set square (for example the left-hand edge if you are right-handed) and hold it firmly so it does not move.
3. Hold the ruler firmly so that it does not move. This makes the set square slide without turning, so the edge that was on AB keeps the same direction — and the line drawn through P along that edge stays parallel to AB.

Constructing Perpendicular Lines

1. Two lines are perpendicular when they meet at a right angle, which is $90\,^{\circ}$.
2. A set square and a ruler.
3. You press firmly on the ruler.
4. They are perpendicular, written $PQ \perp AB$.

Bisecting a line segment

1. To bisect AB means to divide it into two equal halves. The point where the bisecting line PQ crosses AB is its midpoint M, so $AM = MB$.

2. You must keep the same radius — do not change the compass opening. This way the arcs from B cut the arcs from A at P above and Q below, so P and Q are the crossings that the joining line runs through.
3. M is the midpoint of AB, so $AM = MB$, and PQ meets AB at $90\,^\circ$.
4. Each half is 6 cm, and the bisecting line meets PQ at $90\,^\circ$.

Constructing a 90° Angle

1. a. Mark point C on the line AB.
   b. With compasses at C, draw an arc cutting the line at two points D and E.
   c. Open the compasses wider than DC, and with the point at D draw an arc above the line.
   d. With the same wider setting at E, draw another arc cutting the first at F.
   e. Join C to F with a ruler.
   f. CF is perpendicular to AB, so $\angle FCB = 90\,^\circ$.

   Follow the six steps of the construction exactly, ending with the perpendicular CF.
2. With the original width DC, the arcs drawn from D and E would only meet again back at C itself, so you would not reach the new point F. Opening wider lets the two arcs meet at a point F above the line, which is what joins to C to make the perpendicular.

   The step-2 width is exactly the distance from C to each of D and E, so arcs of that width centred at D and E both pass through C and meet only there. A wider opening makes them cross at a fresh point F above the line.
3. CF is perpendicular to AB, so $\angle FCB = 90\,^\circ$.

   Since CF was drawn perpendicular to AB, the angle at the foot of the perpendicular is a right angle, $90\,^\circ$.

Constructing a 60° Angle

1. Your finished figure should show a line segment with B at the left, an arc from B cutting the line at X, a second arc from X (same radius) cutting the first at A, and a line from B to A. The angle between BA and the original line is 60°.

   Draw the line and mark B at its left end and C at its right. Set your compasses to any convenient radius. With the point on B, swing an arc that cuts the line at X. Without changing the radius, put the point on X and swing another arc so it crosses the first; mark that crossing A. Join B to A. Because BX, XA and BA are all equal, triangle BAX is equilateral and the angle at B is 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-2fac1792	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____